Advanced Internet Protocols, Services, and Applications

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PREFACE

Today, the Internet and computer networking are essential to business, learning, personal communication, and entertainment. The infrastructure that carries virtually all messages and transactions sent over the Internet is based on advanced Internet protocols. These advanced Internet protocols ensure that both public and private networks operate with maximum performance, security, and flexibility.

This book is intended to provide a comprehensive technical overview and survey of advanced Internet protocols. First, a solid introduction and discussion of internetworking technologies, architectures, and protocols is provided. The book also presents the application of these concepts into next-generation networks and discusses protection and restoration as well as various tunneling protocols and applications. Finally, emerging topics are discussed.

This book covers basic concepts in Transmission Control Protocol (TCP)/Internet Protocol (IP), Internet architecture, IP routing protocols including transport-layer protocols, IP version 6, Multiprotocol Label Switching (MPLS), networking services such as IP Quality of Service (QoS), IP Multicast, anycast, Layer-2 and Layer-3 virtual private networks (L2VPN and L3VPN), and applications such as IP over Dense Wavelength Division Multiplexing (DWDM), IP traffic engineering, IP in mobility, and IP network security.

Although there are no strict prerequisites for reading this book, a data communication background would be helpful. All the concepts in this book are developed from basics on an intuitive basis, with further insight provided through examples of real-world networks, services, and applications. The authors are well-experienced researchers and engineers from both industry and academia.

Dr. Mallikarjun Tatipamula first offered a course on Advanced Internet Protocols to graduate students at a number of leading universities in addition to tutorials at various conferences. The positive response from students triggered the idea of writing this book with the objective of setting a strong foundation for students regarding Internet protocols.

The authors have developed some of the contents of this book in their graduate courses and seminars in their universities and organizations. These contents have been improved, thanks to feedback from students and industry colleagues. The courses in which this material has been used have attracted both academic and industrial practitioners, as the Internet and computer networking are key topics in the information technology industry. These are people interested in the principles of the Internet and advanced networking technologies, including designing networks and network elements and being able to consult network designers in order to satisfy their customers' needs.

Audience

This book is intended for graduate students, R&D managers, software and hardware engineers, system engineers, and telecommunications and networking professionals. This book will interest those who are currently active or anticipate future involvement in internetworking and are seeking a broad understanding and comprehensive technical overview of Internet technologies, architectures, and protocols including current status and future direction.

Organization

The book is organized as follows.

- Chapter 1 describes Transmission Control Protocol (TCP)/Internet Protocol (IP), which is an Internet protocol stack that enables communications between two computers, or hosts, through the Internet. It is a collection of different protocols. A protocol is a set of rules that controls the way data is sent between hosts.
- Chapter 2 explains protocols in the transport layer, which is the fourth layer of the Open Systems Interconnection (OSI) reference model. Transparent transfer of data between end users using services of the network layer is provided. The well-known protocols in this layer are TCP, User Datagram Protocol (UDP), Stream Control Transmission Protocol (SCTP), and Real-time Transport Protocol (RTP).
- Chapter 3 describes Internet architecture, including basic Internet topology, Internet exchange points (IXPs), the history of IXPs, and the principles of Internet relationships and Internet service providers (ISPs).
- Chapter 4 describes IP routing protocol including an overview of Interior Gateway Protocols (IGPs) and Exterior Gateway Protocols (EGPs). Routing Information Protocol (RIP) and Open Shortest Path First (OSPF) is covered as an IGP, and Boarder Gateway Protocol (BGP) is covered as an EGP.
- Chapter 5 describes Multiprotocol Label Switching (MPLS) technologies, which enable networks to perform traffic engineering, resident communications, virtual private networks, Ethernet emulation, and so on. First, an overview to MPLS is given. Next, the functions and mechanisms of MPLS are described. Finally, MPLS applicabilities are discussed.

- Chapter 6 presents a myriad of concepts and paradigms used to provision quality of service by the Internet. It includes discussions on the implementation of mechanisms of traffic differentiators and policy and traffic policers and regulators. The discussion includes Diffserv, IntServ, and a combination of both as developed by recent research.
- Chapter 7 describes IP multicast and anycast. The majority of the traffic on the Internet is *unicast*: one source device sending to one destination device. However, where the number of receivers is more than one, multicast traffic can be transmitted over the Internet in ways that avoid loading the network with excessive traffic. The multicast mechanisms present the possibility of having multiple receivers for routing and packet delivery.
- Chapter 8 presents several Layer-2 encapsulation protocols. These encapsulation protocols are used to allow Layer-2 connectivity through nonadjacent networks. These protocols allow remote networks to communicate as if they were in the same network. Several examples of these encapsulation protocols are presented including Martini, Layer-2 Transport, and Pseudowire Emulation Edge to Edge protocol.
- Chapter 9 introduces point-to-point virtual connectivity and virtual broadcast access connectivity, such as virtual local area networks (LANs). Layer-2 virtual private network over transport can provide several features such as redundancy against failures and controlled connectivity through a packet-switched network or Generalized MPLS (GMPLS). Multiple Layer-2 access protocols can be virtually emulated.
- Chapter 10 introduces the multilayer network evolution, followed by discussion on limitations in current legacy networks, automated provisioning in IP/optical networks, the proposed control plane techniques in the industry, key design requirements for next-generation multilayer IP/optical networks, and comparisons of the proposed control plane models against these key design requirements.
- Chapter 11 describes the main differences between IPv4 and IPv6, investigates the reasons for the lagging uptake of IPv6 so far, and explains why, contrary to widespread belief, IPv6's large address space alone is insufficient to drive early IPv6 deployment.
- Chapter 12 describes several routing schemes to maximize the network utilization for various traffic demand models for MPLS and OSPF technologies. One useful approach to enhancing routing performance is to minimize the maximum link utilization rate, also called the network congestion ratio, of all network links. Minimizing the network congestion ratio leads to an increase in admissible traffic.
- Chapter 13 discusses some popular threats and how they exploit protocol vulnerabilities. Some counter measures, based on additional algorithms that either work as an application or as a protocol at the network or transport layers, are discussed including methods to trace back threatening hosts.

- Chapter 14 explains the different ways of augmenting the Internet for mobility support, as well as the security threats that need to be considered as part of this. The relationship between mobility and multihoming will be explored further. The chapter also gives an overview of auxiliary protocols that play a fundamental role in IP mobility management.

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Christian Vogt

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Several chapters of the book are based on our research works. We would like to thank the people who have contributed materials to some chapters, especially Dr. Nattapong Kitsuwan (UEC Tokyo), Mohammad Kamrul Islam (UEC Tokyo), Khondaker M. Salehin (New Jersey Institute of Technology), and Tuhina Sarkar (New Jersey Institute of Technology).

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Eiji Oki is an Associate Professor at the University of Electro-Communications, Tokyo, Japan. He received his Bachelor and Master of Engineering degrees in Instrumentation Engineering and a Doctorate of Philosophy in Electrical Engineering from Keio University, Yokohama, Japan, in 1991, 1993, and 1999, respectively. In 1993, he joined Nippon Telegraph and Telephone Corporation (NTT) Communication Switching Laboratories, Tokyo, Japan. He has been researching network design and control, traffic-control methods, and high-speed switching systems. From 2000 to 2001, he was a Visiting Scholar at the Polytechnic Institute of New York University, Brooklyn, New York, where he was involved in designing terabit switch/router systems. He was engaged in researching and developing high-speed optical IP backbone networks with NTT Laboratories. He joined the University of Electro-Communications, Tokyo, Japan, in July 2008. He has been active in standardization of path computation element (PCE) and GMPLS in the Internet Engineering Task Force (IETF). He wrote 11 IETF RFCs. He served as a guest co-editor for the special issue on "Multi-Domain Optical Networks: Issues and Challenges," June 2008, in IEEE Communications Magazine; a guest co-editor for the special issue on Routing, "Path Computation and Traffic Engineering in Future Internet," December 2007, in the Journal of Communications and Networks; a guest co-editor for the special section on "Photonic Network Technologies in Terabit Network Era," April 2011, in IEICE Transactions on Communications; a Technical Program Committee (TPC) Co-Chair for the Workshop on High-Performance Switching and Routing in 2006 and 2010; a Track Co-Chair on Optical Networking for ICCCN 2009; a TPC Co-Chair for the International Conference on IP+Optical Network (iPOP 2010); and a Co-Chair of Optical Networks and Systems Symposium for IEEE ICC 2011. Professor Oki was the recipient of the 1998 Switching System Research Award and the 1999 Excellent Paper Award presented by IEICE, the 2001 Asia-Pacific Outstanding Young Researcher Award presented by IEEE Communications Society for his contribution to broadband network, ATM, and optical IP technologies, and the 2010 Telecom System Technology Prize by the Telecommunications Advanced Foundation. He has co-authored two books, *Broadband Packet Switching Technologies*, published by John Wiley & Sons, Inc., New York, in 2001, and *GMPLS Technologies*, published by CRC Press, Boca Raton, Florida, in 2005. He is an IEEE Senior Member.

Roberto Rojas-Cessa received a Master of Computer Engineering degree and a Doctorate of Philosophy in Electrical Engineering from the Polytechnic Institute of New York University, Brooklyn, New York. He also received a Master of Science in Electrical Engineering from the Research and Advanced Studies Center (CIVESTAV) in Mexico. He received his Bachelor of Science in Electronic Instrumentation from Universidad Veracruzana, Mexico. Currently, he is an Associate Professor in the Department of Electrical and Computer Engineering, New Jersey Institute of Technology, Newark, New Jersey. He was an Adjunct Professor and a Research Associate in the Department of Electrical and Computer Engineering of Polytechnic Institute of New York University. He has been involved in design and implementation of application-specific integrated-circuits (ASIC) for biomedical applications and high-speed computer communications, and in the development of high-performance and scalable packet switches and reliable switches. He was part of the team designing a 40 Tb/s core router at Coree, Inc. in Tinton Falls, NJ. His research interests include high-speed switching and routing, fault tolerance, quality-of-service networks, network measurements, and distributed systems. His research has been funded by the U.S. National Science Foundation and Industry. He was the recipient of the Advance in Research Excellence of the ECE Department in 2004. He has served on several technical committees for IEEE conferences and as a reviewer for several IEEE journals. He has been a reviewer and panelist for the U.S. National Science Foundation and the U.S. Department of Energy. He has more than 10 years of experience in teaching Internet protocols and computer communications. Currently, he is the Director of the Networking Research Laboratory at the ECE Department and the Coordinator of the Networking Research Focus Area Group of the same department.

Mallikarjun Tatipamula is Head of Packet Technologies Research, Ericsson Silicon Valley. He leads a research team and is responsible for innovation and implementation of leading-edge technologies including Openflow, next-generation routing architectures, application-aware networking, and Cloud computing, networking, and services. He closely works with leaders from universities, research and education networks, and service providers around the world. Prior to his role at Ericsson, he was Vice President and Head of Service Provider Sector at Juniper Networks, Sunnyvale, California. His team responsibilities include new technologies, architectures, standards, solutions, and creation of business strategy for the implementation of next-generation products in content delivery network/video, Cloud, IP/optical integration, security, mobility, and convergence. Prior to Juniper, he was with Cisco systems for over eight years and made active contributions to the Cisco IP NGN strategy. These contributions include “Service Exchange Framework”, IMS/FMC, advanced technologies such as IPv6, GMPLS, multicast, along with his contributions in the early days of VoIP, mobile wireless and IP/optical integration that led to definition and implementation plans. Prior to Cisco, Mallikarjun was a principal engineer at Motorola in the Cellular Infrastructure Group. He was responsible for defining system architecture for advanced wireless and satellite systems. From 1993 to 1997 he was a senior member of the scientific staff at BNR (now Nortel), Ottawa, responsible for development of Nortel's optical (OC12/OC48) products and involved in the development of the Nortel CDMA Base Station. From 1990 to 1992, Mallikarjun was with Indian Telephone Industries as an Assistant Executive Engineer in Optical Transmission R&D Laboratories and Indian Institute of Technology, Chennai, as a Senior Project Officer in the Fiber Optics Labs, responsible for development of optical data links. He is a key note speaker at leading telecommunications and networking events. Mallikarjun obtained Doctorate of Philosophy in Information Science and Technology from the University of Tokyo, Japan, a Master of Science in Communication Systems and High Frequency Technologies from the Indian Institute of Technology, Chennai, and a Bachelor of Technology in Electronics and Communication Engineering from NIT, Warangal, India. Mallikarjun has authored or coauthored a number of patents and publications with leaders from NRENs and service providers. He is a trusted partner, advisor, and thought leader. Mallikarjun is a lead editor for "Multimedia Communication Networks: Technologies and Services," by Artech House Publishers. He is a senior member of IEEE and his biography appeared in Marquis Who is Who in the World, Who is Who in America, and Who is Who Science and Engineering. Mallikarjun delivered distinguished lectures at leading universities including Stanford University, the University of Tokyo, the Tokyo Institute of Technology, Tsing Hua University, Beijing University, China, and IIT Delhi.

Christian Vogt is a Senior Marketing Manager at Ericsson Silicon Valley. He works on product and marketing strategies regarding the convergence of wireless and wireline networks, and their transition from IP version 4 to 6. His technical interests further include Internet routing and addressing, IP mobility and multihoming, related security aspects, as well as next-generation Internet architectures. As part of this work, Christian co-chairs the Internet Area and Source Address Validation Improvements working groups in the Internet Engineering Task Force. Christian received his doctoral degree from the University of Karlsruhe in Germany in 2007 for his dissertation on efficient and secure mobility support in IP version 6. He also holds a Master of Science in Computer Science from the University of Southern California, Los Angeles, and a German Diplom in Computer Science from the University of Bonn. Currently, Christian is pursuing an Executive Master of Business Administration degree at the Wharton School, University of Pennsylvania.

1

TRANSMISSION CONTROL PROTOCOL/INTERNET PROTOCOL OVERVIEW

This first chapter provides an overview of Transmission Control Protocol (TCP)/Internet Protocol (IP), which is an Internet protocol stack to perform communications between two computers, or hosts, through the Internet. It is a collection of different protocols. A *protocol* is a set of rules that controls the way data is transmitted between hosts.

1.1 FUNDAMENTAL ARCHITECTURE

Figure 1.1 shows the Open Systems Interconnection (OSI) reference model and the Internet protocol stack. The International Standardization Organization (ISO) specifies a guideline called the OSI reference model. It is an abstract description for layered communications and computer network protocol design. It consists of seven layers, which are, from the bottom, physical, data link, network, transport, session, presentation, and application, as shown in Figure 1.1**a**. The application layer is the OSI layer closest to the end user, which means that both the OSI application layer and the user interact directly with the software application. This layer interacts with software applications that implement a communicating component. At the physical layer, data is recognized, or handled, as bits. At the data link layer, data is handled as frames. At the network layer, data is recognized as packets. In the transport layer, data is handled as segments or datagrams. In the remaining upper layers, users' information is recognized.

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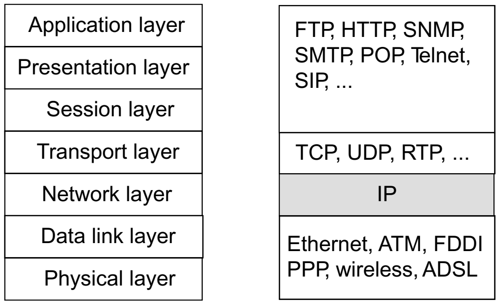

(a) OSI reference model (b) Internet protocol stack

Figure 1.1 Layer model.

The Internet protocol stack is shown in Figure 1.1**b**, where we can see to which layer in the OSI reference model each protocol corresponds. For example, the Internet protocol corresponds to the network layer. TCP corresponds to the transport layer. The Internet protocol stack that includes several protocols, as shown in Figure 1.1**b**, is referred to as *TCP/IP*.

TCP/IP is being standardized by the Internet Engineering Task Force (IETF). It is widely used as the *de facto* standard protocol for building network equipment and internetworking. If the TCP/IP standard is used, computers can communicate with each other regardless of hardware and operating system.

Figure 1.2 shows a basic philosophy of TCP/IP. In the TCP/IP philosophy, the IP core network functions simply and quickly, while the functionality of the edges surrounding the core is complex. Edges can be hosts, edge routers, network boundaries, etc. The IP network based on this philosophy is scalable and flexible. This enables different complexities at the edge.

Application data and customer traffic can be transmitted on the IP layer because IP is the least common denominator, as shown in Figure 1.3. IP supports transport-layer protocols such as TCP, User Datagram Protocol (UDP), Real Time Protocol (RTP)/UDP, and Stream Control Transmission Protocol (SCTP), as shown in

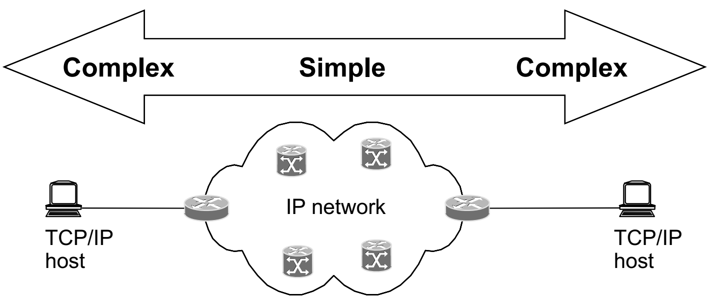

Figure 1.2 Basic philosophy of TCP/IP.

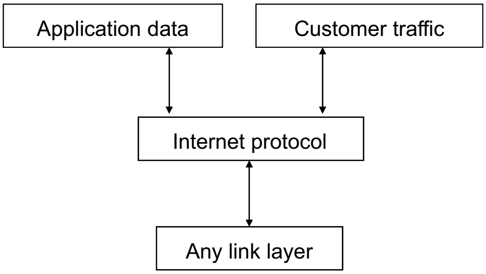

Figure 1.3 IP is the least common denominator.

Figure 1.4. IP works on link-layer protocols such as Ethernet, Point-to-Point Protocol (PPP) over Asynchronous Transer Mode (ATM), satellite, wireless, optical, and IP/Multiprotocol Label Switching (MPLS), as shown in Figure 1.5.

More than one network can be connected via IP, as shown in Figure 1.6. IP enables building a large-scale network where smaller networks, or subnetworks, are concatenated to build one large network. Inside each network, a node does not need to be connected with other nodes in the same network via IP. This is the way the Internet is built.

Figure 1.7 shows the protocol stack based on the OSI reference model and network equipment and protocols connecting layers. A repeater is a functional device of layer 1. It enhances signal level. A bridge, or switch, processes frames, for example Ethernet frames, corresponding to layer 2. Two bridges are connected via Ethernet protocol. A

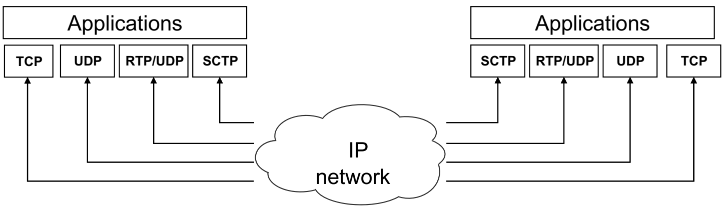

Figure 1.4 Any transport-layer protocols on IP.

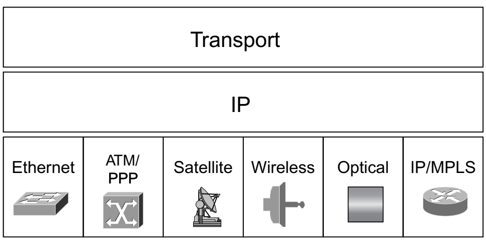

Figure 1.5 Any link-layer protocols under IP.

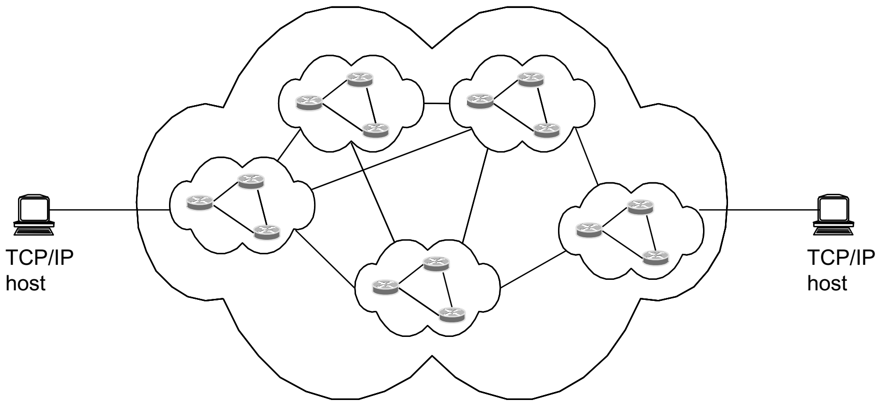

Figure 1.6 Interworking via IP.

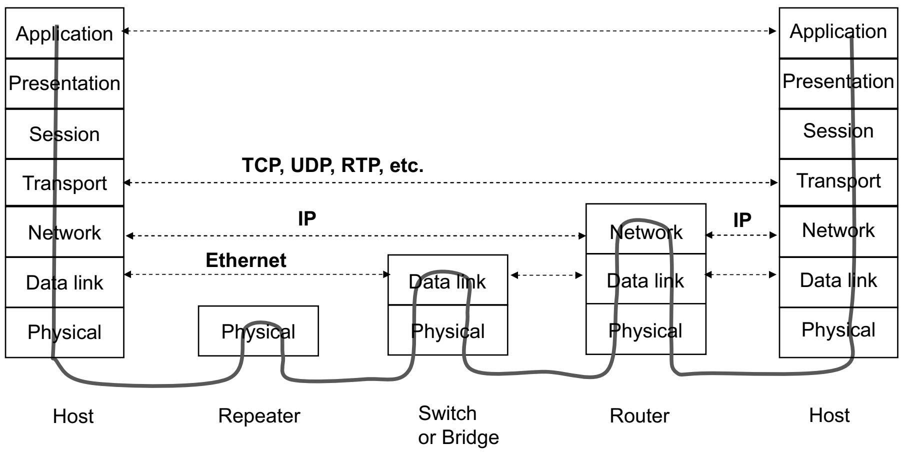

Figure 1.7 Protocol stack and connectivity.

router processes IP packets corresponding to layer 3. Two routers are connected via IP. Two hosts are connected via protocols for layers 4, 5, 6, and 7. For example, TCP, UDP, and RPT are used to connect two hosts on layer 4.

1.2 INTERNET PROTOCOL BASICS

Internet Protocol is a protocol in the network layer. It is used for sending data from a source host to its destination host where each host has a unique number. The IP header format is shown in Figure 1.8. Version specifies the IP version. Header length indicates the size of the header. Type of service is used to guide the selection of the